

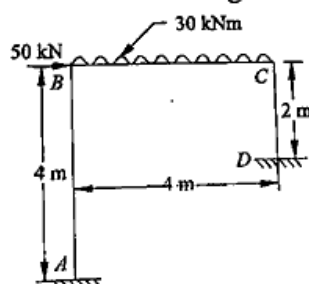
Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

	(Level/CO)	Marks
Q. 1 Solve <u>any two</u> of the following.		12
A) A uniformly distributed load of 15 kN/m covering a length of 3 m crosses a girder of span 10 m. Find the maximum shear force and bending moment at a section 4 m from L. H. support.	Understand / CO3	06
B) Determine the bending moment developed anywhere on the girder of span 15 m due to two rolling loads 150 kN and 100 kN spaced 4 m apart with the 100 kN load leading passing over the girder. Find the equivalent UDL to give the same maximum bending moment.	Apply / CO3	06
C) A uniformly distributed load of 40 kN/m longer than the span rolls over a girder of 30 m span. Using Influence Line diagram for shear force (SF) and bending moment (BM) determine the maximum SF & BM at a section 12 m from left hand support A.	Analyze / CO3	06
Q.2 Solve <u>any two</u> following questions.		12
A) A three hinged arch of span 40 m and rise 8 m carries a concentrated loads of 200 kN and 150 kN at a distance of 8 m and 16 m from the left end and a uniformly distributed load of 50 kN/m on the right half of the span. Find the horizontal thrust .	Understand	06
B) A suspension bridge is of 50 m span with a 16 m wide roadway. It is subjected to a load of 25 kN/m ² including dead loads. The bridge is supported by a pair of cables having a central dip of 4.2 m. Find the cross-sectional area of the cable if the maximum permissible stress in the cable material is not to exceed 600 N/mm ² .	Apply	06
C) A suspension bridge of 250 m span has two nos. of three hinged stiffening girders supported by cables with a central dip of 25 m. If 4-point loads of 300 kN each are placed at the centre line of the roadway at 20, 30, 40 and 50 m from the left-hand hinge, find the shear force and bending moment in each girder at 62.5 m from each end. Calculate also the maximum tension in the cable.	Analyze	06

Q. 3 Solve any two of the following. 12

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| A) State and explain the steps for analysis of indeterminate structures by direct flexibility method . | Understand/
CO1 | 06 |
| B) Two span continuous beam ABC of span AB 4 m & span BC 3 m with Support A as fixed, Support B (middle support) and C as roller Support. Span AB is subjected to UDL of 60 kN/m. Span BC is subjected 100 kN at distance 1.5 m from C. Draw SFD and BMD . Use direct flexibility method . | Apply /
CO1 | 06 |
| C) Analyse the portal frame ABCD as shown in the figure below by flexibility matrix method . | Analyze /
CO1 | 06 |



Q.4 Solve any two of the following. 12

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|--|---------------------|----|
| A) State and explain the steps for analysis of indeterminate structures by direct stiffness method. | Understand/
CO1 | 06 |
| B) Two span continuous beam ABC of span AB 4 m & span BC 3 m with Support A as fixed, Support B (middle support) and C as roller Support. Span AB is subjected to UDL of 60 kN/m. Span BC is subjected 100 kN at distance 1.5 m from C. Draw SFD and BMD . Use direct stiffness method (i. e. Displacement method) | Apply /
CO1 | 06 |
| C) Explain in detail the finite difference method. | Understand
/ CO2 | 06 |

Q. 5 Solve any two of the following. 12

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| A) What is mean by 'Discretisation' in finite element analysis. List the different elements & guidelines used for Discretisation of any complex geometry and explain them in detail. | Understand/
CO2 | 06 |
| B) What are the pros and cons of Finite Element Method. | Understand/
CO2 | 06 |
| C) Derive shape function for bar element. | Apply/
CO2 | 06 |

*** End ***